

Lab Problem 5

Objective:

To implement **Locally Weighted Linear Regression (LWLR)** and analyze how the bandwidth parameter τ controls model behavior and the bias–variance trade-off.

Dataset: Use the **Diabetes dataset** from `scikit-learn`.

- Use **only one input feature: BMI**
- Use the **disease progression score** as the target variable
- **Normalize the input feature** to zero mean and unit variance before applying LWLR

Tasks:

You must perform the following tasks in order:

Task 1: Baseline Model (Global Regression)

- Load the Diabetes dataset and extract the BMI feature.
- Normalize the feature.
- **Fit global polynomial regression models of degree 4 and degree 7** to the data.
- Plot the data points along with both polynomial fits on the same figure.
- Compare the two models and briefly comment on their limitations in capturing local variations in the data.

Task 2: Weighting Function

- Implement a Gaussian weighting function: `get_weights(x_q, X,  $\tau$ )`
- The function should return a **diagonal weight matrix** where weights decrease with the distance between each training point and the query point x_q .

Task 3: Mathematical Framework

- For each query point x_q , LWLR constructs a **local linear model** by minimizing the **weighted squared error**:

$$E(x_q) = \frac{1}{2} \sum_i w_i (y_i - \hat{y}_i)^2$$

- where the weights are defined using a **Gaussian kernel**:

$$w_i = \exp\left(-\left(x^{(i)} - x_q\right)^2 / (2\tau^2)\right)$$

- The locally optimal parameter vector is computed using the **weighted normal equation**:

$$\theta(x_q) = (X^T W X)^{-1} X^T W y$$

Use this formulation to compute predictions at each query point.

Task 4: Bandwidth Analysis

Using the **normalized input**, perform LWLR for the following bandwidth values:

- $\tau = 0.05$
 - $\tau = 0.30$
 - $\tau = 1.20$
1. Plot the LWLR prediction curves for all three values of τ on the same figure.
 2. **Identify which curve corresponds to:**
 - a. Overfitting
 - b. Optimal fitting
 - c. Underfitting